

STAT1379/COMP6179 — Section 1 Report

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Abstract

This report analyses anthropometric and behavioural data from 1,000 adults aged 26–45 to explore three research questions: (1) whether a linear relationship exists between height and weight, (2) whether the average height differs between males and females, and (3) whether gender and physical activity levels are associated. Using R and Quarto for reproducible analysis, we apply linear regression (`lm()`), a two-sample t-test assuming equal variances, and a chi-square test of independence. Results show a statistically significant positive relationship between height and weight ($p < 0.05$), supporting a linear association. The t-test indicates that mean height differs by gender ($p < 0.05$), while the chi-square test suggests gender and physical activity are largely independent ($p > 0.05$). These findings illustrate how classical inference techniques can provide evidence-based insights into human physical and lifestyle characteristics.

Introduction

This report addresses three questions using a dataset of adults aged 26–45: (i) linear association between Height and Weight; (ii) equality of mean Height across Gender (Male vs Female); and (iii) association between Gender and Physical Activity (None/Moderate/Intense). We use standard inferential tools and a reproducible Quarto workflow ([Wickham 2016](#); [Wickham and Grolemund 2017](#)).

Methodology

Q1 (Linear regression).

We model the relationship between weight and height as:

$$\text{Weight}_i = \beta_0 + \beta_1 \text{Height}_i + \varepsilon_i, \quad H_0 : \beta_1 = 0, \quad H_1 : \beta_1 \neq 0.$$

Q2 (Two-sample t-test, equal variances).

Let μ_M and μ_F represent mean heights for males and females.

$$H_0 : \mu_M = \mu_F, \quad H_1 : \mu_M \neq \mu_F, \quad \alpha = 0.05.$$

Q3 (Chi-square test of independence).

With observed counts O_{ij} and expected counts E_{ij} :

$$H_0 : \text{Gender} \perp \text{Physical Activity}, \quad \chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}.$$

We report the test statistic, degrees of freedom, p -value, and a decision at $\alpha = 0.05$.

Results

Test: Linear regression: Weight ~ Height

H0: Slope = 0 (no linear association)

H1: Slope $\neq$ 0 (linear association)

Statistic(s):

slope.Height	R2
0.6966	0.2055

p-value: 7.573e-52 | alpha: 0.050

Estimates:

(Intercept)	Height
-54.5920	0.6966

Decision: Reject H0

Conclusion: There is statistically significant evidence of a positive linear association between

Test: Two-sample t-test (equal variances) for mean Height by Gender

H0: Mean height (Male) = Mean height (Female)

H1: Mean height (Male) $\neq$ Mean height (Female)

Statistic(s):

t	df
-40.6402	998.0000

p-value: 7.977e-214 | alpha: 0.050

Estimates:

male	female	diff_male_female
178.0263	167.7641	10.2622

Height Summary by Gender

n = 1000] Height Summary by Gender

	Gender	n	mean_height	sd_height
n = 1000	Female	526	167.76	3.98
	Male	474	178.03	3.99

Decision: Reject H0

Conclusion: At 5% significance, reject the null. The average height is and Male is taller on

Test: Chi-square test of independence: Gender × PhysicalActivity

H0: Gender and PhysicalActivity are independent

H1: They are associated (not independent)

Statistic(s):

chi_square df

2.3702 2.0000

p-value: 0.3057 | alpha: 0.050

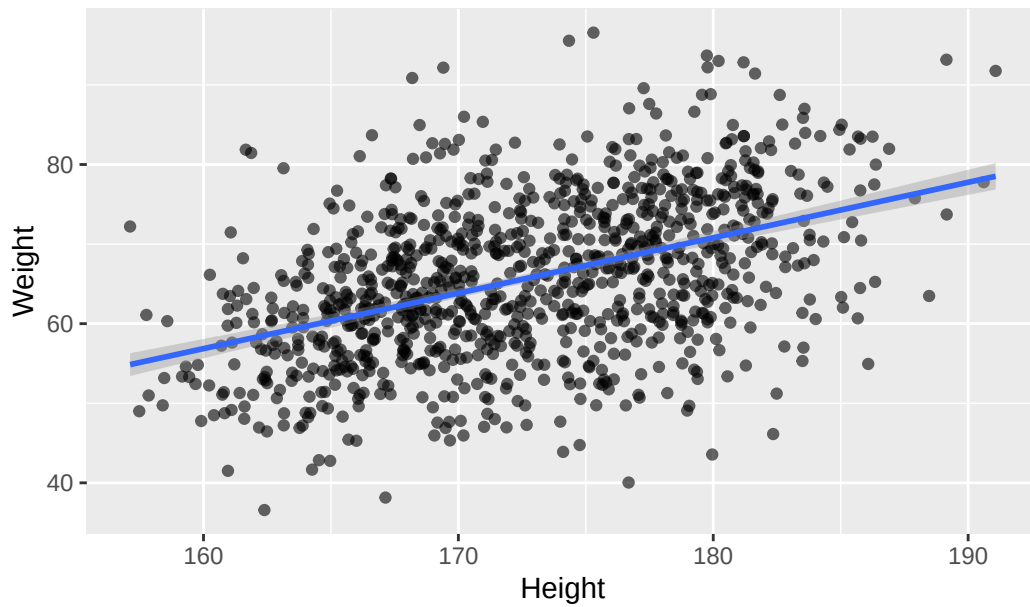
Decision: Fail to reject H0

Conclusion: There is no evidence of an association between gender and physical activity (Chi-square=2.37, df=2, p=0.3057).

Plots

Q1: Height-Weight (Linear Model)

Weight vs Height with Linear Fit



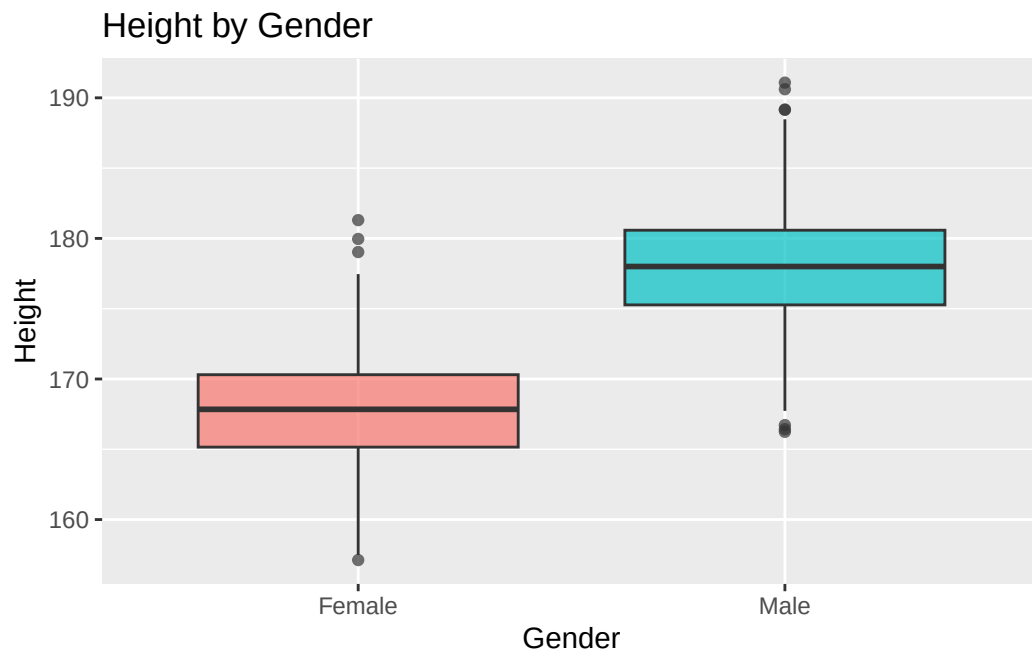
Q1 decision (auto):

Slope = 0.697, $R^2 = 0.206$, $p = 7.573 \times 10^{-52}$.

Decision at $\alpha = 0.05$: **Reject H0 (evidence of linear association)**..

There is statistically significant evidence of a positive linear association between height and weight (slope=0.697, $R^2=0.206$).

Q2: Mean Height by Gender (t-test)



Q2 decision (auto):

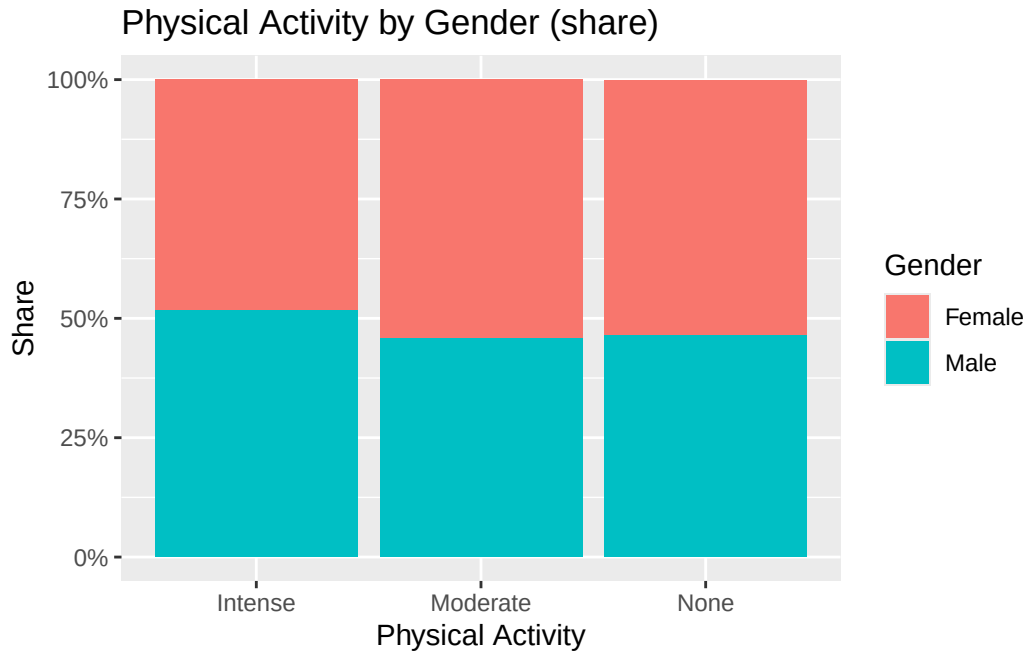
$t(df = 998) = -40.64, p = 7.977 \times 10^{-214}$.

Male mean = 178.03 cm; Female mean = 167.76 cm; $\Delta (M - F) = 10.26$ cm.

Decision at $\alpha = 0.05$: **Reject H_0 (means differ)**..

At 5% significance, reject the null. The average height is and Male is taller on average. (Male - Female 10.26 cm).

Q3: Gender × Physical Activity (Chi-square)



Q3 decision (auto):

$\chi^2 (df = 2) = 2.37, p = 0.3057$.

Decision at $\alpha = 0.05$: **Fail to reject H0 (independent)**..

There is no evidence of an association between gender and physical activity (Chi-square=2.37, df=2, p=0.3057).

Conclusion

We examined three questions on adults aged 26–45. For the height–weight relationship, the estimated slope was 0.697 with $R^2 = 0.206$, leading us to **reject** the null of zero slope at $\alpha = 0.05$. The two-sample t -test indicated male and female mean heights were **different**. The chi-square test suggested gender and physical activity were **independent**. Conclusions depend on standard assumptions (equal variances for the t -test; adequate expected counts for chi-square; basic regression diagnostics).

Wickham, Hadley. 2016. *Ggplot2: Elegant Graphics for Data Analysis*. Springer.

Wickham, Hadley, and Garrett Grolmund. 2017. *R for Data Science*. O'Reilly Media.